

Digital Signal Processing

Lecture

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Discrete-Time Fourier Series

- For periodic sequences: $\tilde{x}[n] = \tilde{x}[n + kN], \forall k$

- Discrete Fourier Series representation:

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi}{N}kn}$$

- Only N terms: there are only N different frequencies of period N
- definition: $W_N = e^{-j\frac{2\pi}{N}}$

$$\tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j2\pi \frac{kn}{N}} \quad \text{inverse DFS}$$

Proof:

$$\begin{aligned} \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j2\pi \frac{kn}{N}} &= \sum_{n=0}^{N-1} \left\{ \frac{1}{N} \sum_{p=0}^{N-1} \tilde{X}(p) e^{j2\pi \frac{pn}{N}} \right\} e^{-j2\pi \frac{kn}{N}} \\ &= \sum_{p=0}^{N-1} \tilde{X}(p) \underbrace{\left\{ \frac{1}{N} \sum_{n=0}^{N-1} e^{j2\pi \frac{(p-k)n}{N}} \right\}}_{\delta(p-k)} = \tilde{X}(k) \end{aligned}$$

$$\tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j2\pi \frac{kn}{N}} \quad \text{analysis}$$

$$\tilde{x}(n) = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}(k) e^{j2\pi \frac{kn}{N}} \quad \text{synthesis}$$

The conventional (continuous-time) FS represent a periodic signal using an infinite number of complex exponentials, whereas the DFS represent such a signal using a finite number of complex exponentials

Example 1

- DFS of a periodic impulse train

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} \delta[n - rN] = \begin{cases} 1 & n = rN \\ 0 & \text{else} \end{cases}$$

- Since the period of the signal is N

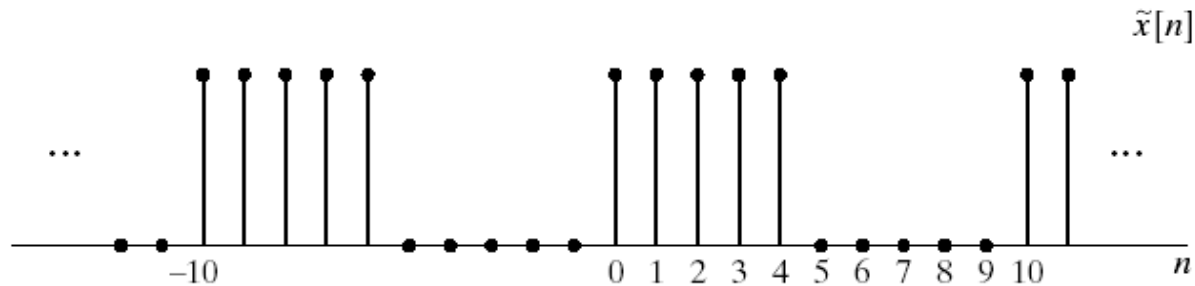
$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn} = \sum_{n=0}^{N-1} \delta[n] e^{-j(2\pi/N)kn} = e^{-j(2\pi/N)k0} = 1$$

- We can represent the signal with the DFS coefficients as

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} \delta[n - rN] = \frac{1}{N} \sum_{k=0}^{N-1} e^{j(2\pi/N)kn}$$

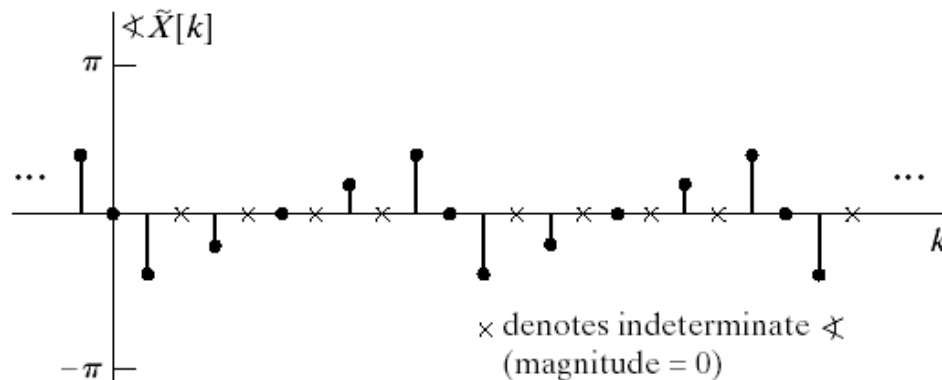
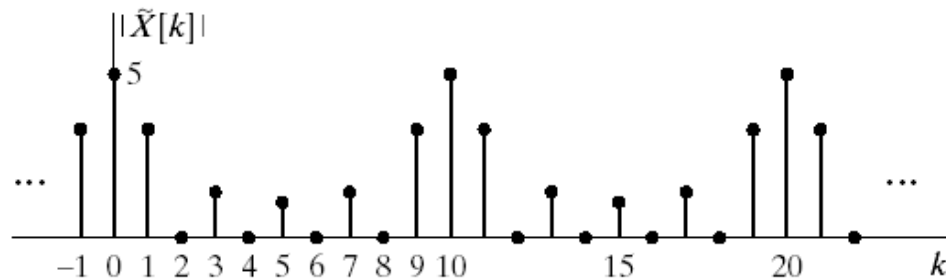
Example 2

- DFS of an periodic rectangular pulse train



- The DFS coefficients

$$\tilde{X}[k] = \sum_{n=0}^4 e^{-j(2\pi/10)kn} = \frac{1 - e^{-j(2\pi/10)k5}}{1 - e^{-j(2\pi/10)k}} = e^{-j(4\pi k/10)} \frac{\sin(\pi k/2)}{\sin(\pi k/10)}$$



Properties of DFS

- Linearity

$$\begin{array}{ccc}
 \tilde{x}_1[n] & \xleftrightarrow{\text{DFS}} & \tilde{X}_1[k] \\
 \tilde{x}_2[n] & \xleftrightarrow{\text{DFS}} & \tilde{X}_2[k] \\
 a\tilde{x}_1[n] + b\tilde{x}_2[n] & \xleftrightarrow{\text{DFS}} & a\tilde{X}_1[k] + b\tilde{X}_2[k]
 \end{array}$$

- Shift of a Sequence

$$\begin{array}{ccc}
 \tilde{x}[n] & \xleftrightarrow{\text{DFS}} & \tilde{X}[k] \\
 \tilde{x}[n - m] & \xleftrightarrow{\text{DFS}} & e^{-j2\pi km / N} \tilde{X}[k] \\
 e^{j2\pi nm / N} \tilde{x}[n] & \xleftrightarrow{\text{DFS}} & \tilde{X}[k - m]
 \end{array}$$

- Duality

$$\begin{array}{ccc}
 \tilde{x}[n] & \xleftrightarrow{\text{DFS}} & \tilde{X}[k] \\
 \tilde{X}[n] & \xleftrightarrow{\text{DFS}} & N\tilde{x}[-k]
 \end{array}$$

Symmetry Properties

Periodic Sequence (Period N)	DFS Coefficients (Period N)
1. $\tilde{x}[n]$	$\tilde{X}[k]$ periodic with period N
2. $\tilde{x}_1[n], \tilde{x}_2[n]$	$\tilde{X}_1[k], \tilde{X}_2[k]$ periodic with period N
3. $a\tilde{x}_1[n] + b\tilde{x}_2[n]$	$a\tilde{X}_1[k] + b\tilde{X}_2[k]$
4. $\tilde{X}[n]$	$N\tilde{x}[-k]$
5. $\tilde{x}[n - m]$	$W_N^{km} \tilde{X}[k]$
6. $W_N^{-\ell n} \tilde{x}[n]$	$\tilde{X}[k - \ell]$
7. $\sum_{m=0}^{N-1} \tilde{x}_1[m] \tilde{x}_2[n - m]$ (periodic convolution)	$\tilde{X}_1[k] \tilde{X}_2[k]$
8. $\tilde{x}_1[n] \tilde{x}_2[n]$	$\frac{1}{N} \sum_{\ell=0}^{N-1} \tilde{X}_1[\ell] \tilde{X}_2[k - \ell]$ (periodic convolution)
9. $\tilde{x}^*[n]$	$\tilde{X}^*[-k]$

Symmetry Properties Cont'd

Periodic Sequence (Period N)	DFS Coefficients (Period N)
10. $\tilde{x}^*[-n]$	$\tilde{X}^*[k]$
11. $\mathcal{Re}\{\tilde{x}[n]\}$	$\tilde{X}_e[k] = \frac{1}{2}(\tilde{X}[k] + \tilde{X}^*[-k])$
12. $j\mathcal{Im}\{\tilde{x}[n]\}$	$\tilde{X}_o[k] = \frac{1}{2}(\tilde{X}[k] - \tilde{X}^*[-k])$
13. $\tilde{x}_e[n] = \frac{1}{2}(\tilde{x}[n] + \tilde{x}^*[-n])$	$\mathcal{Re}\{\tilde{X}[k]\}$
14. $\tilde{x}_o[n] = \frac{1}{2}(\tilde{x}[n] - \tilde{x}^*[-n])$	$j\mathcal{Im}\{\tilde{X}[k]\}$
Properties 15–17 apply only when $x[n]$ is real.	
15. Symmetry properties for $\tilde{x}[n]$ real.	$\left\{ \begin{array}{l} \tilde{X}[k] = \tilde{X}^*[-k] \\ \mathcal{Re}\{\tilde{X}[k]\} = \mathcal{Re}\{\tilde{X}[-k]\} \\ \mathcal{Im}\{\tilde{X}[k]\} = -\mathcal{Im}\{\tilde{X}[-k]\} \\ \tilde{X}[k] = \tilde{X}[-k] \\ \angle \tilde{X}[k] = -\angle \tilde{X}[-k] \end{array} \right.$
16. $\tilde{x}_e[n] = \frac{1}{2}(\tilde{x}[n] + \tilde{x}[-n])$	$\mathcal{Re}\{\tilde{X}[k]\}$
17. $\tilde{x}_o[n] = \frac{1}{2}(\tilde{x}[n] - \tilde{x}[-n])$	$j\mathcal{Im}\{\tilde{X}[k]\}$

Periodic Convolution

- Take two periodic sequences

$$\begin{array}{ccc} \tilde{x}_1[n] & \xleftrightarrow{\text{DFS}} & \tilde{X}_1[k] \\ \tilde{x}_2[n] & \xleftrightarrow{\text{DFS}} & \tilde{X}_2[k] \end{array}$$

- Let's form the product

$$\tilde{X}_3[k] = \tilde{X}_1[k] \tilde{X}_2[k]$$

- The periodic sequence with given DFS can be written as

$$\tilde{x}_3[n] = \sum_{m=0}^{N-1} \tilde{x}_1[m] \tilde{x}_2[n-m]$$

- Periodic convolution is commutative

$$\tilde{x}_3[n] = \sum_{m=0}^{N-1} \tilde{x}_2[m] \tilde{x}_1[n-m]$$

Periodic Convolution Cont'd

$$\tilde{x}_3[n] = \sum_{m=0}^{N-1} \tilde{x}_1[m] \tilde{x}_2[n-m]$$

- Substitute periodic convolution into the DFS equation

$$\tilde{X}_3[k] = \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} \tilde{x}_1[m] \tilde{x}_2[n-m] \right) W_N^{kn}$$

- Interchange summations

$$\tilde{X}_3[k] = \sum_{m=0}^{N-1} \tilde{x}_1[m] \left(\sum_{n=0}^{N-1} \tilde{x}_2[n-m] W_N^{kn} \right)$$

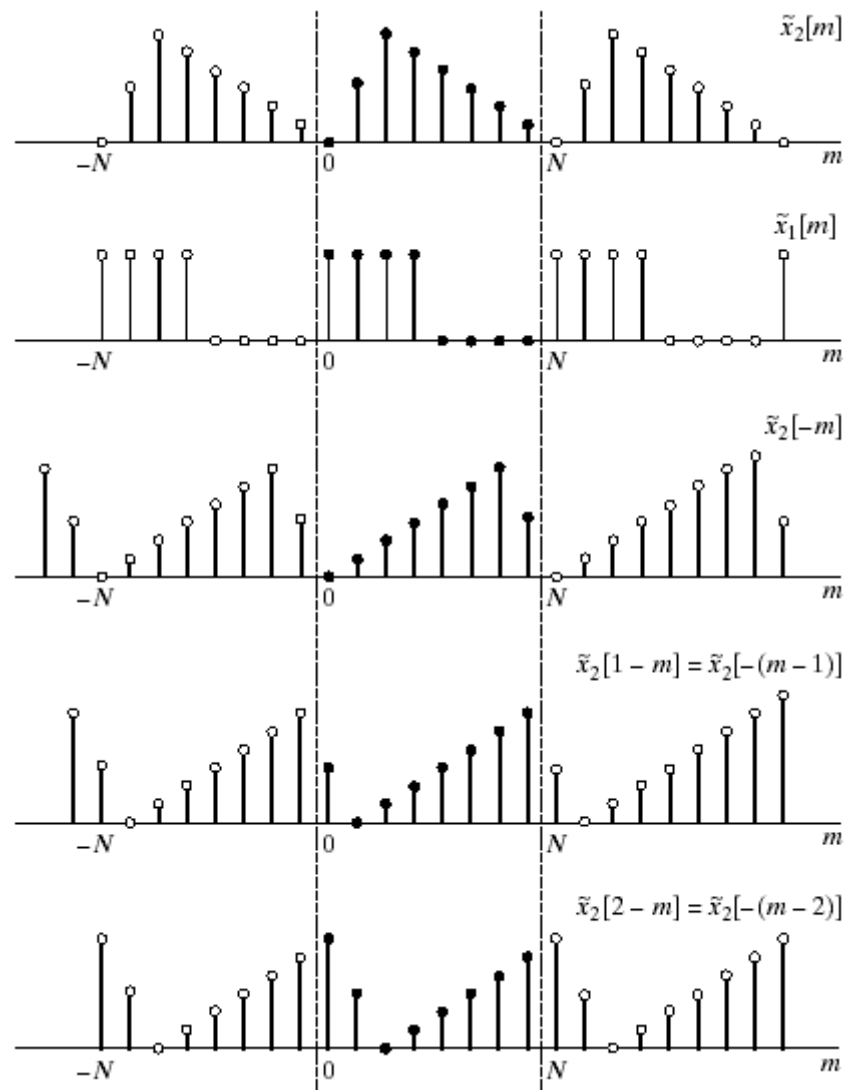
- The inner sum is the DFS of shifted sequence

$$\sum_{n=0}^{N-1} \tilde{x}_2[n-m] W_N^{kn} = W_N^{km} \tilde{X}_2[k]$$

- Substituting

$$\tilde{X}_3[k] = \sum_{m=0}^{N-1} \tilde{x}_1[m] \left(\sum_{n=0}^{N-1} \tilde{x}_2[n-m] W_N^{kn} \right) = \sum_{m=0}^{N-1} \tilde{x}_1[m] W_N^{km} \tilde{X}_2[k] = \tilde{X}_1[k] \tilde{X}_2[k]$$

Graphical Periodic Convolution



DTFT to DFT

Recall the DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

The DTFT is not suitable for a practical DSP because:

- in any DSP application, we are able to compute the spectrum **only at specific discrete values** of ω
- any signal in any DSP application can be measured **only in a finite number of points**

Sampling the Fourier Transform

- Consider an aperiodic sequence with a Fourier transform

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega})$$

- Assume that a sequence is obtained by sampling the DTFT

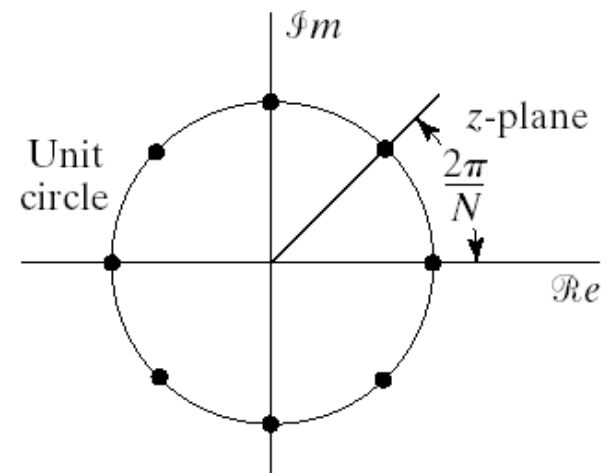
$$\tilde{X}[k] = X(e^{j\omega}) \Big|_{\omega=(2\pi/N)k} = X(e^{j(2\pi/N)k})$$

- Since the DTFT is periodic resulting sequence is also periodic
- We can also write it in terms of the z-transform

$$\tilde{X}[k] = X(z) \Big|_{z=e^{j(2\pi/N)k}} = X(e^{j(2\pi/N)k})$$

- The sampling points are shown in figure
- $\tilde{X}[k]$ could be the DFS of a sequence
- Write the corresponding sequence

$$\tilde{X}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}$$



DFT Analysis and Synthesis

The DFT transform:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi \frac{kn}{N}} \quad \text{analysis}$$

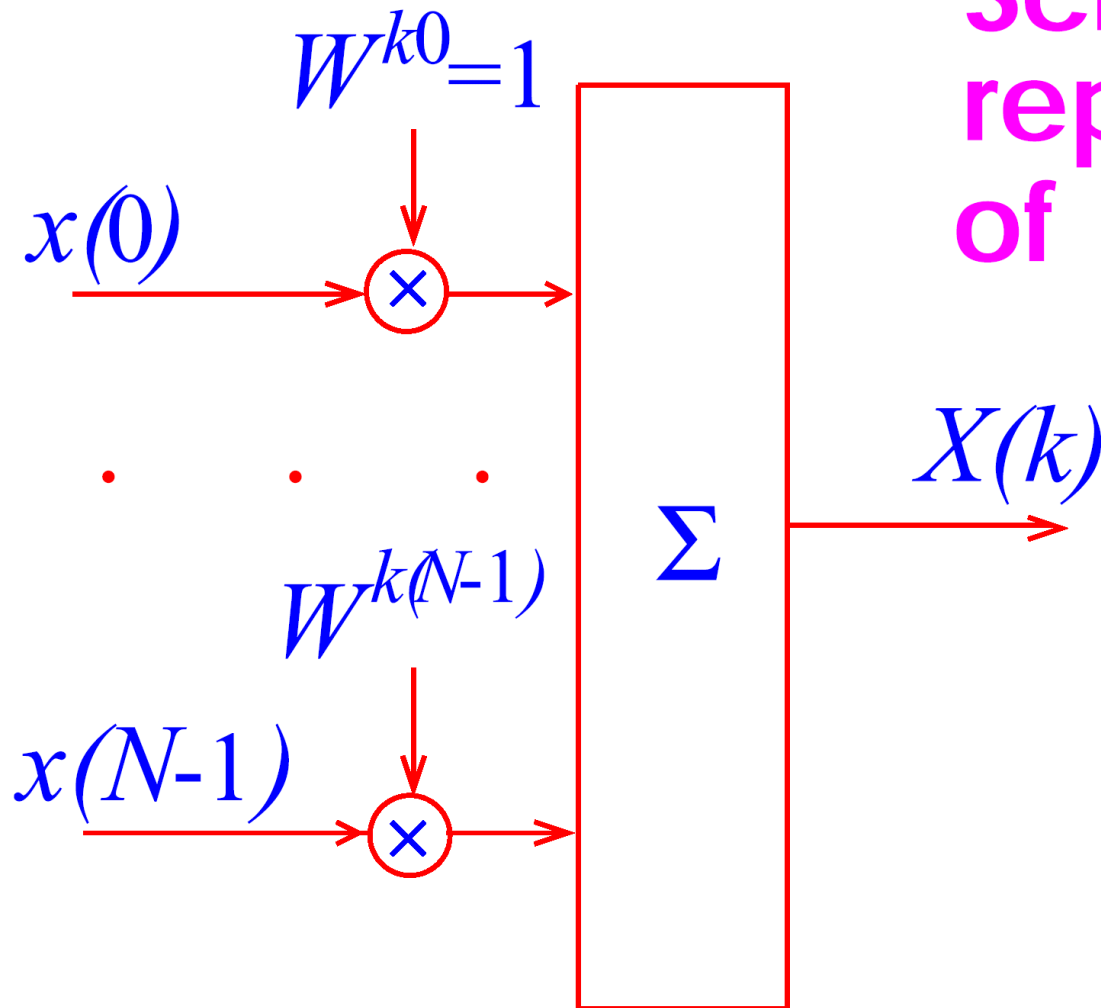
$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi \frac{kn}{N}} \quad \text{synthesis}$$

Alternative formulation:

$$X(k) = \sum_{n=0}^{N-1} x(n) W^{kn} \quad \longleftarrow \quad W = e^{-j\frac{2\pi}{N}}$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W^{-kn}$$

Schematic representation of DFT



DFT is Periodic with period N

$$\begin{aligned} X(k + N) &= \sum_{n=0}^{N-1} x(n) e^{-j2\pi \frac{(k+N)n}{N}} \\ &= \left(\sum_{n=0}^{N-1} x(n) e^{-j2\pi \frac{kn}{N}} \right) e^{-j2\pi n} \\ &= X(k) e^{-j2\pi n} = X(k) \quad \Rightarrow \end{aligned}$$

the DFT-spectrum $X(k)$ is periodic with period N (recall that DTFT-spectrum is periodic as well, but with period 2π)

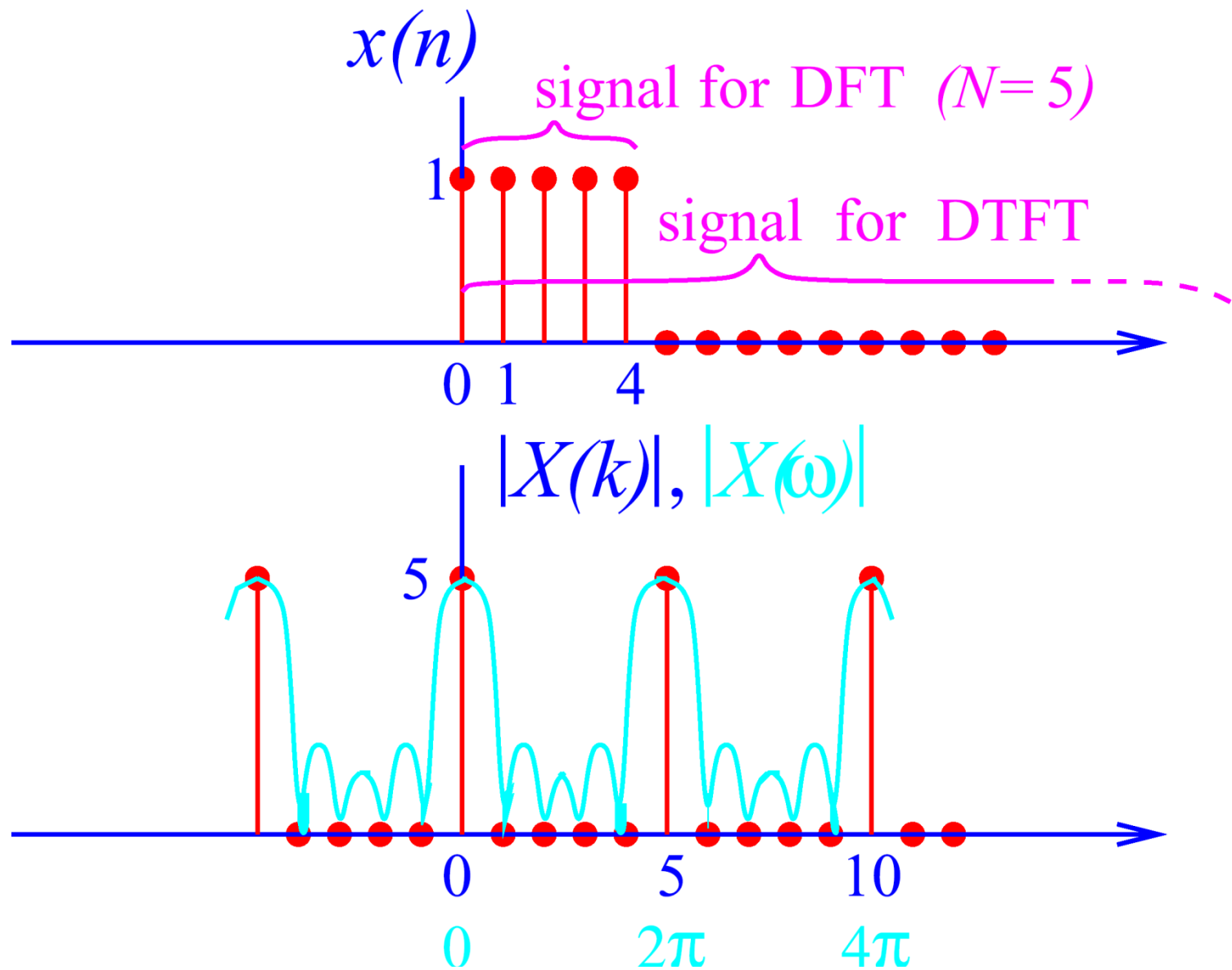
Example 1

$$x(n) = \begin{cases} 1, & 0 \leq n \leq (N - 1) \\ 0, & \text{otherwise} \end{cases}$$

$$X(k) = \sum_{n=0}^{N-1} e^{-j2\pi \frac{kn}{N}} = N \delta(k) \quad \Rightarrow$$

the rectangular pulse is “interpreted” by the DFT as a spectral line at the frequency $\omega = 0$

Example 1 (cont.) $N=5$



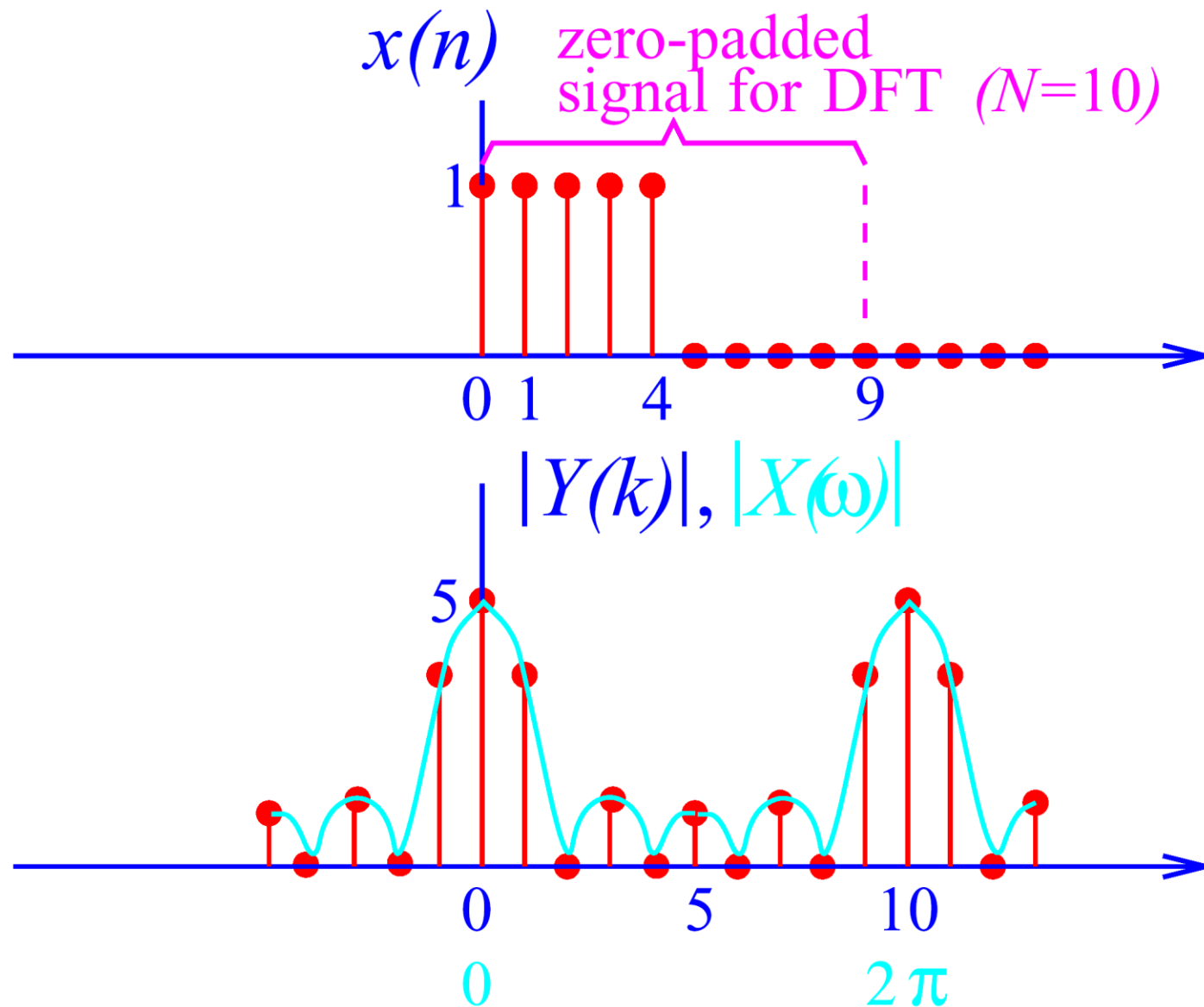
Example 1 (cont.) $N > M$

$$\{y(n)\} = \{x(0), \dots, x(M-1), \underbrace{0, 0, \dots, 0}_{N-M \text{ positions}}\}$$

where $x(0) = \dots = x(M-1) = 1$. Hence, the DFT:

$$\begin{aligned} Y(k) &= \sum_{n=0}^{N-1} y(n) e^{-j2\pi \frac{kn}{N}} = \sum_{n=0}^{M-1} e^{-j2\pi \frac{kn}{N}} \\ &= \frac{\sin\left(\pi \frac{kM}{N}\right)}{\sin\left(\pi \frac{k}{N}\right)} e^{-j\pi \frac{k(M-1)}{N}} \end{aligned}$$

Example 1 (cont.) N=10



DFT: Matrix Form

$$\mathbf{X} = \mathbf{W}\mathbf{x}$$

Result: The inverse DFT is given by

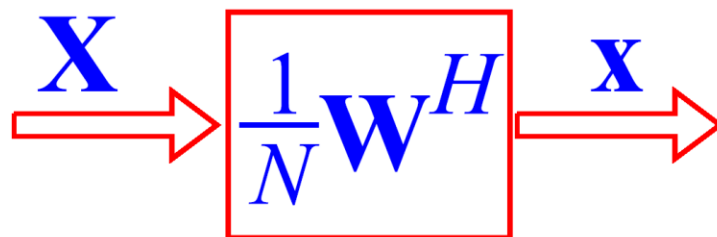
$$\mathbf{x} = \frac{1}{N} \mathbf{W}^H \mathbf{X}$$

Proof: elementary, using the direct checking of the fact that $\mathbf{W}^H \mathbf{W} = \mathbf{W} \mathbf{W}^H = N \mathbf{I}$, where $\mathbf{I}$ is the identity matrix.

DFT



Inverse DFT



DFT from DFS

- a finite-length sequence $x[n]$ can be represented by samples of its DTFT spaced by $\frac{2\pi}{N}$, where $N \geq \text{length}(x[n])$
- these samples are the DFS sequence of a periodized version of $x[n]$

$$\tilde{x}[n] = \sum_{r=-\infty}^{+\infty} x[n - rN] = x[((n))_N] \xleftrightarrow{\mathcal{DFS}} \tilde{X}[k] = X(e^{j\frac{2\pi k}{N}})$$

- If we limit the length of $\tilde{X}[k]$ to one period, we obtain the DFT of $x[n]$:

$$X[k] = \begin{cases} \tilde{X}[k], & 0 \leq k \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

Properties of DFT

Linearity

$$x_1[n] \xleftrightarrow{\text{DFT}} X_1[k]$$

$$x_2[n] \xleftrightarrow{\text{DFT}} X_2[k]$$

$$ax_1[n] + bx_2[n] \xleftrightarrow{\text{DFT}} aX_1[k] + bX_2[k]$$

Duality

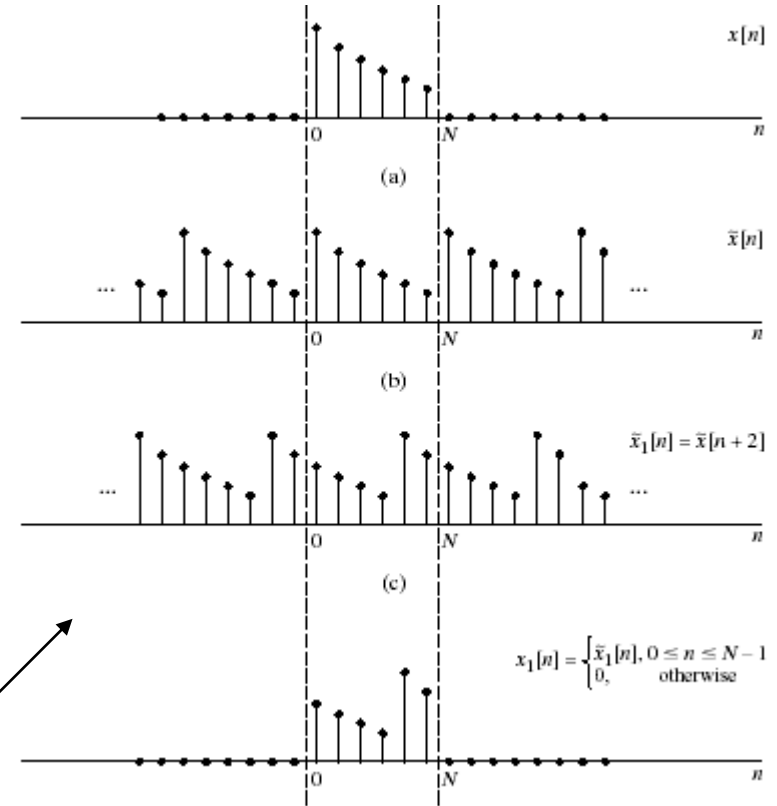
$$x[n] \xleftrightarrow{\text{DFT}} X[k]$$

$$X[n] \xleftrightarrow{\text{DFT}} Nx[((-k))_N]$$

Circular Shift of a Sequence

$$x[n] \xleftrightarrow{\text{DFT}} X[k]$$

$$x[((n-m))_N] \quad 0 \leq n \leq N-1 \xleftrightarrow{\text{DFT}} X[k]e^{-j(2\pi k/N)m}$$



Symmetry Properties

$$5. \quad x[((n-m))_N]$$

$$W_N^{km} X[k]$$

$$6. \quad W_N^{-\ell n} x[n]$$

$$X[((k-\ell))_N]$$

$$7. \quad \sum_{m=0}^{N-1} x_1(m) x_2[((n-m))_N]$$

$$X_1[k] X_2[k]$$

$$8. \quad x_1[n] x_2[n]$$

$$\frac{1}{N} \sum_{\ell=0}^{N-1} X_1(\ell) X_2[((k-\ell))_N]$$

$$9. \quad x^*[n]$$

$$X^*[((-k))_N]$$

$$10. \quad x^*[((-n))_N]$$

$$X^*[k]$$

$$11. \quad \mathcal{Re}\{x[n]\}$$

$$X_{\text{ep}}[k] = \frac{1}{2} \{X[((k))_N] + X^*[(((-k))_N)]\}$$

$$12. \quad j\mathcal{Im}\{x[n]\}$$

$$X_{\text{op}}[k] = \frac{1}{2} \{X[((k))_N] - X^*[(((-k))_N)]\}$$

$$13. \quad x_{\text{ep}}[n] = \frac{1}{2} \{x[n] + x^*[(((-n))_N)]\}$$

$$\mathcal{Re}\{X[k]\}$$

$$14. \quad x_{\text{op}}[n] = \frac{1}{2} \{x[n] - x^*[(((-n))_N)]\}$$

$$j\mathcal{Im}\{X[k]\}$$

Properties 15–17 apply only when $x[n]$ is real.

$$15. \quad \text{Symmetry properties}$$

$$\begin{cases} X[k] = X^*[(((-k))_N)] \\ \mathcal{Re}\{X[k]\} = \mathcal{Re}\{X[(((-k))_N)]\} \\ \mathcal{Im}\{X[k]\} = -\mathcal{Im}\{X[(((-k))_N)]\} \\ |X[k]| = |X[(((-k))_N)]| \\ \angle\{X[k]\} = -\angle\{X[(((-k))_N)]\} \end{cases}$$

DFT Properties

- linearity:

$$ax[n] + by[n] \xleftrightarrow{\mathcal{DFT}} aX[k] + bY[k],$$

where $x[n]$ and $y[n]$ have the same length (after zero-padding if necessary)

- Shift: the sequence that has $W_N^{km}X[k]$ as DFT ?

Let $Y[k] = W_N^{km}X[k]$, then

$$\begin{aligned}\tilde{y}[n] = y[((n))_N] &\xleftrightarrow{\mathcal{DFT}} \tilde{Y}[k] = Y[((k))_N] \\ &= W_N^{((k))_N m} X[((k))_N]\end{aligned}$$

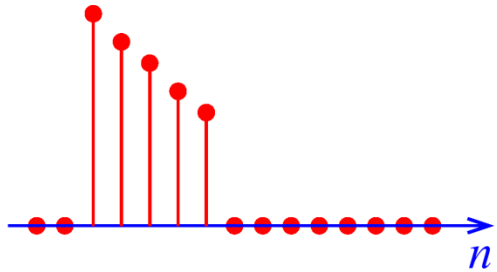
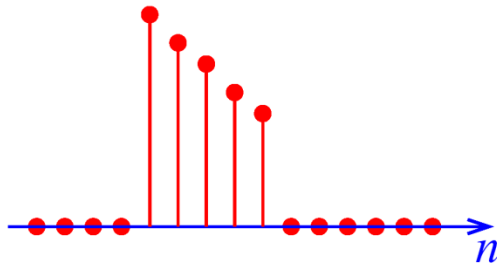
so that $\tilde{y}[n] = \tilde{x}[n - m] = x[((n - m))_N]$ and

$$x[((n - m))_N], 0 \leq n \leq N - 1 \xleftrightarrow{\mathcal{DFT}} W_N^{km} X[k],$$

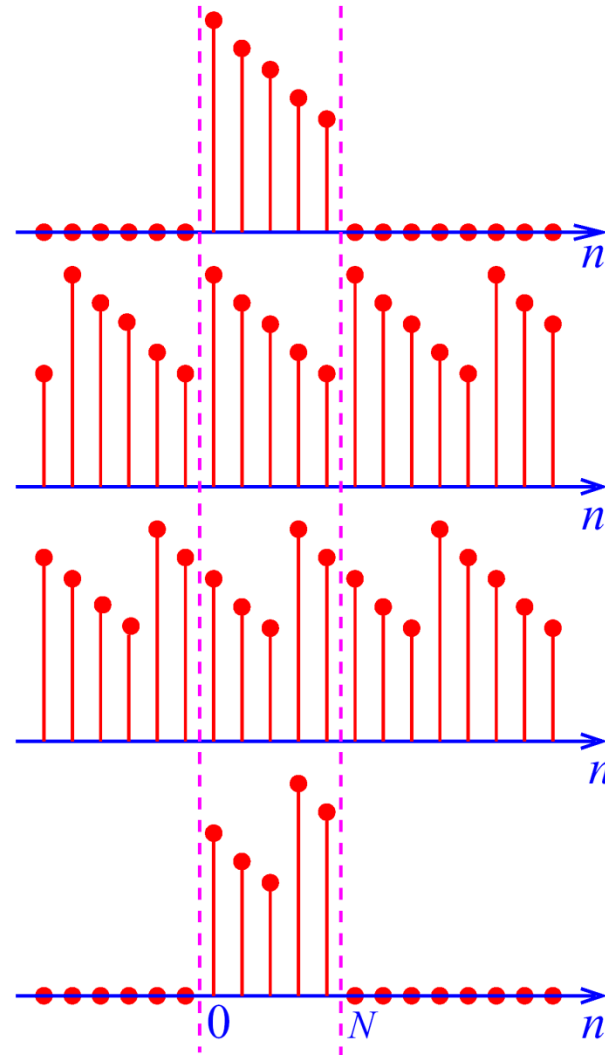
i.e. a *circular shift*

Example: Circular Shift

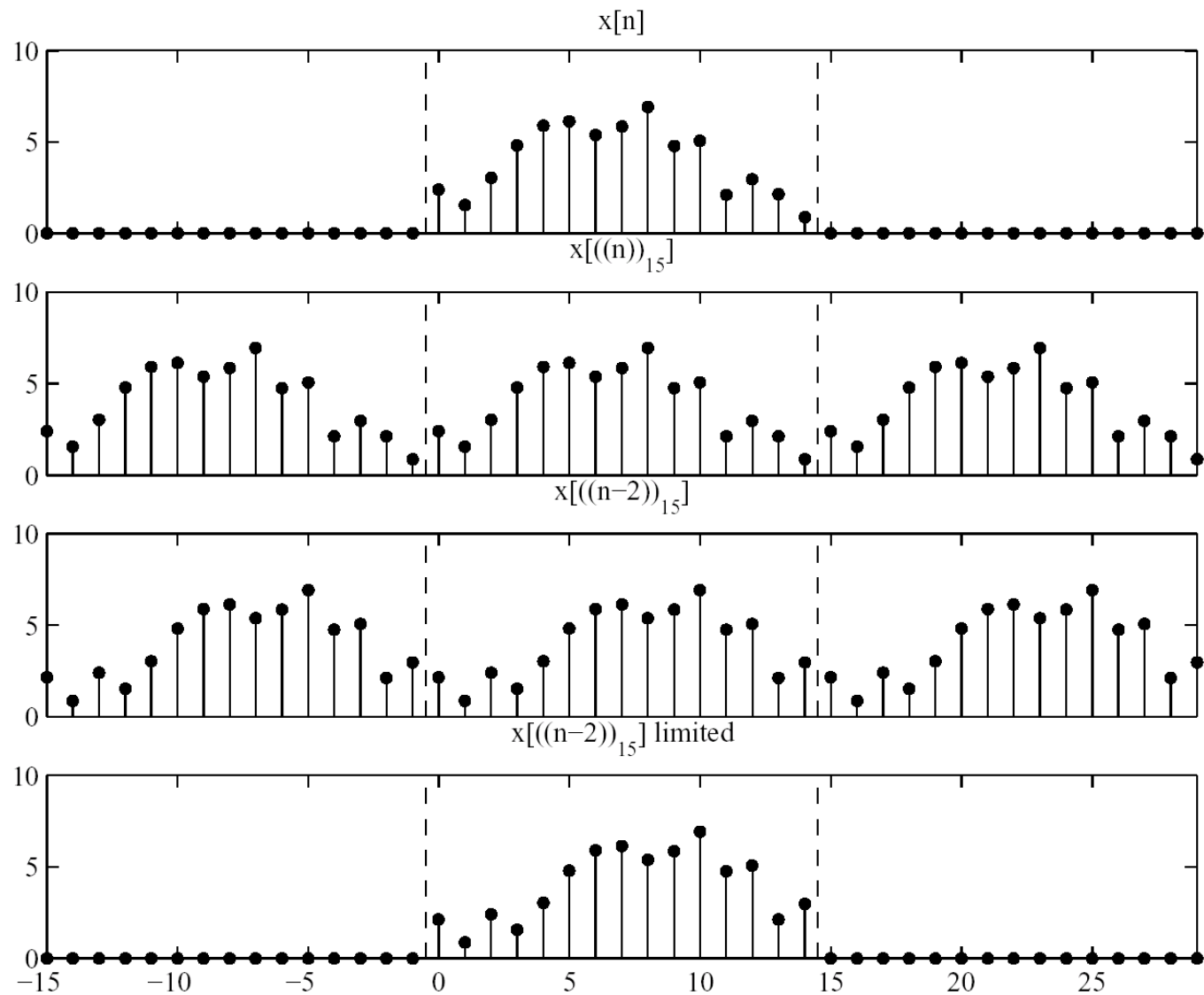
conventional shift



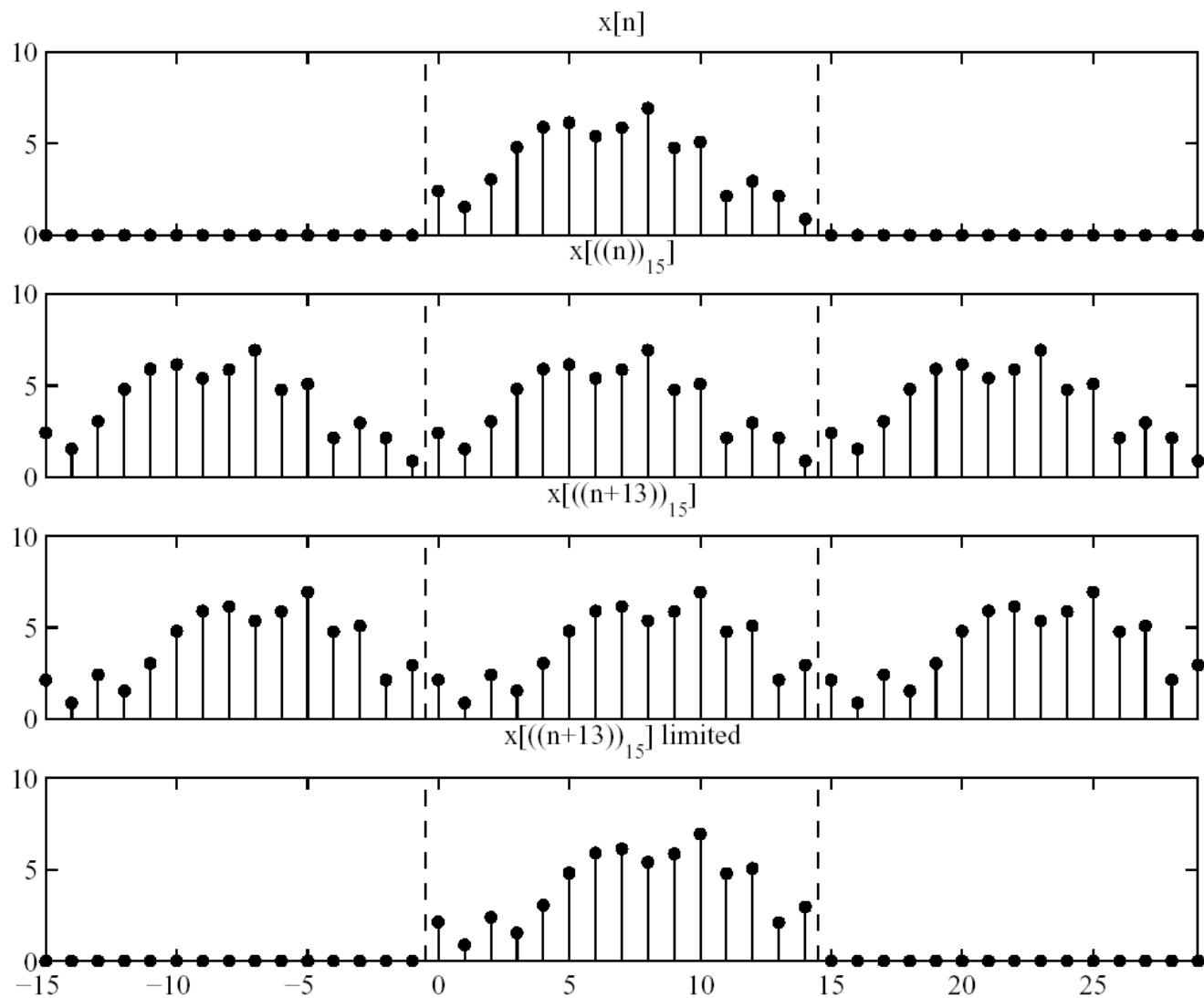
circular shift



Example: Circular Shift



Example: Circular Shift



Duality

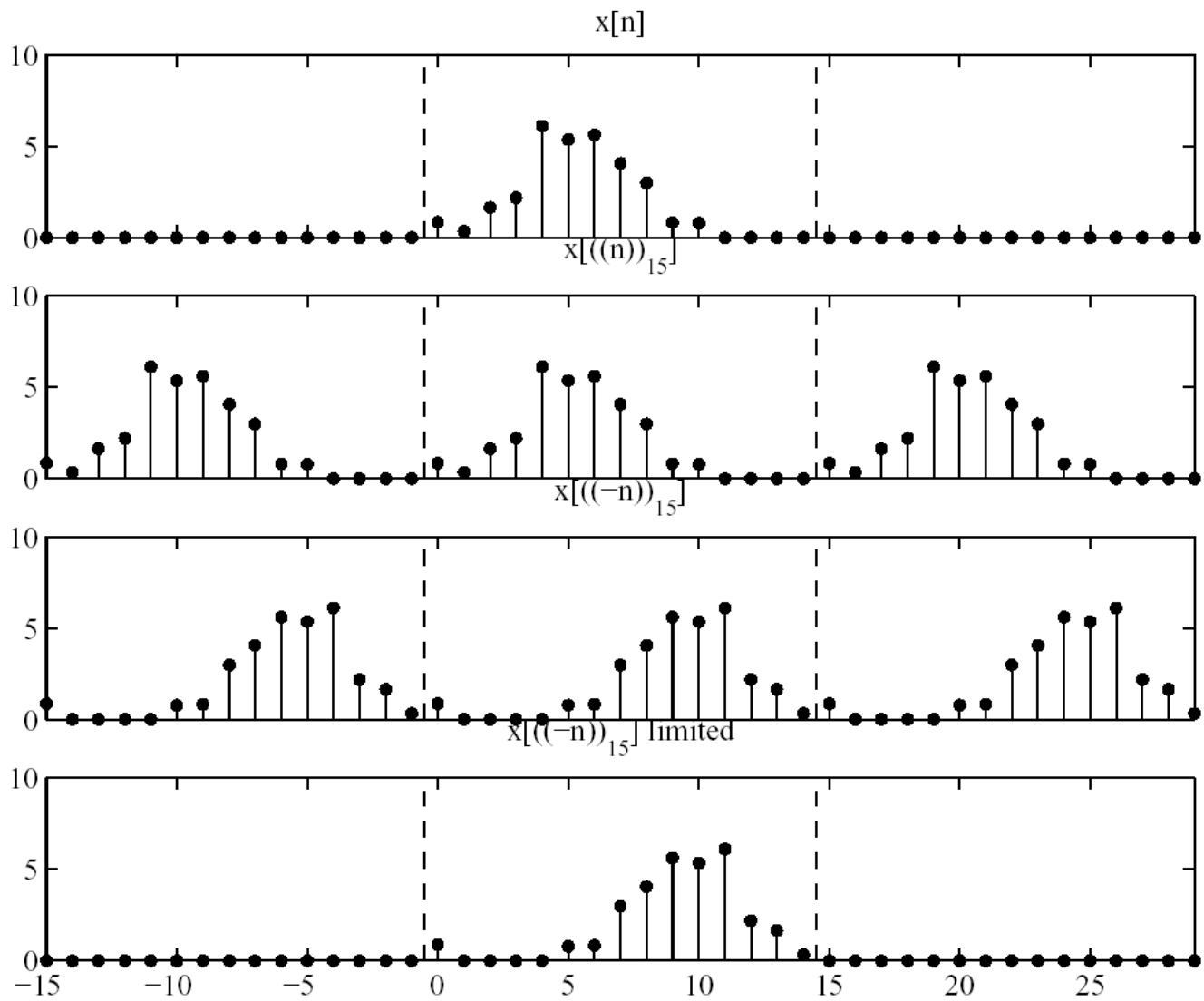
- by using the duality property of the DFS, one proves that

$$X[n] \xleftrightarrow{\mathcal{DFT}} Nx[((-k))_N], 0 \leq k \leq N-1$$

i.e.

- periodize $x[n]$
- flip the horizontal axis
- keep one period from 0 to $N-1$

Circular Flip



Properties: Circular Convolution

- sequence for which $X[k]Y[k]$ is the DFT ? $\Rightarrow$ One period of the periodic sequence

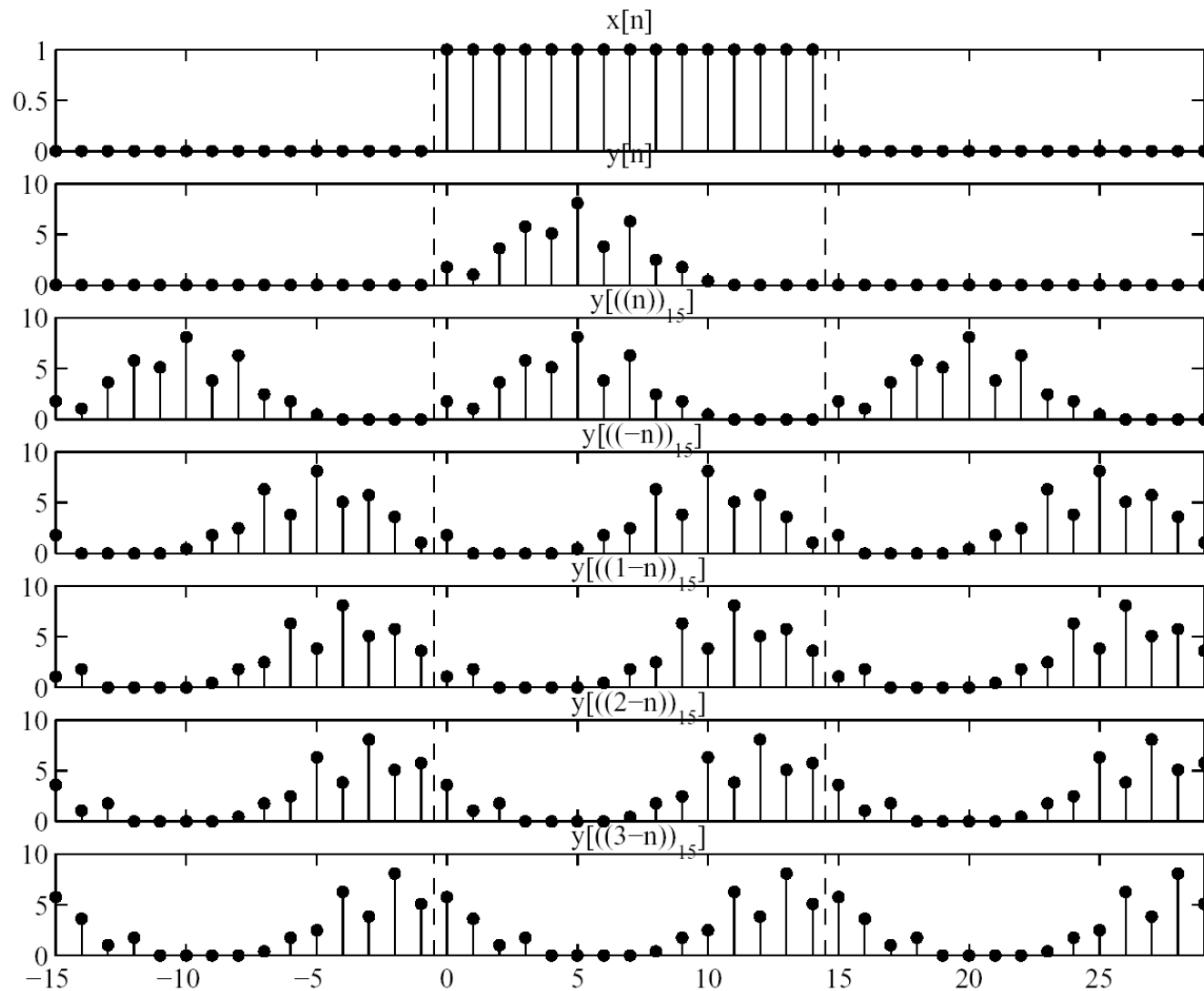
$$\sum_{m=0}^{N-1} \tilde{x}[m]\tilde{y}[n-m] = \sum_{m=0}^{N-1} x[((m))_N]y[((n-m))_N],$$

or

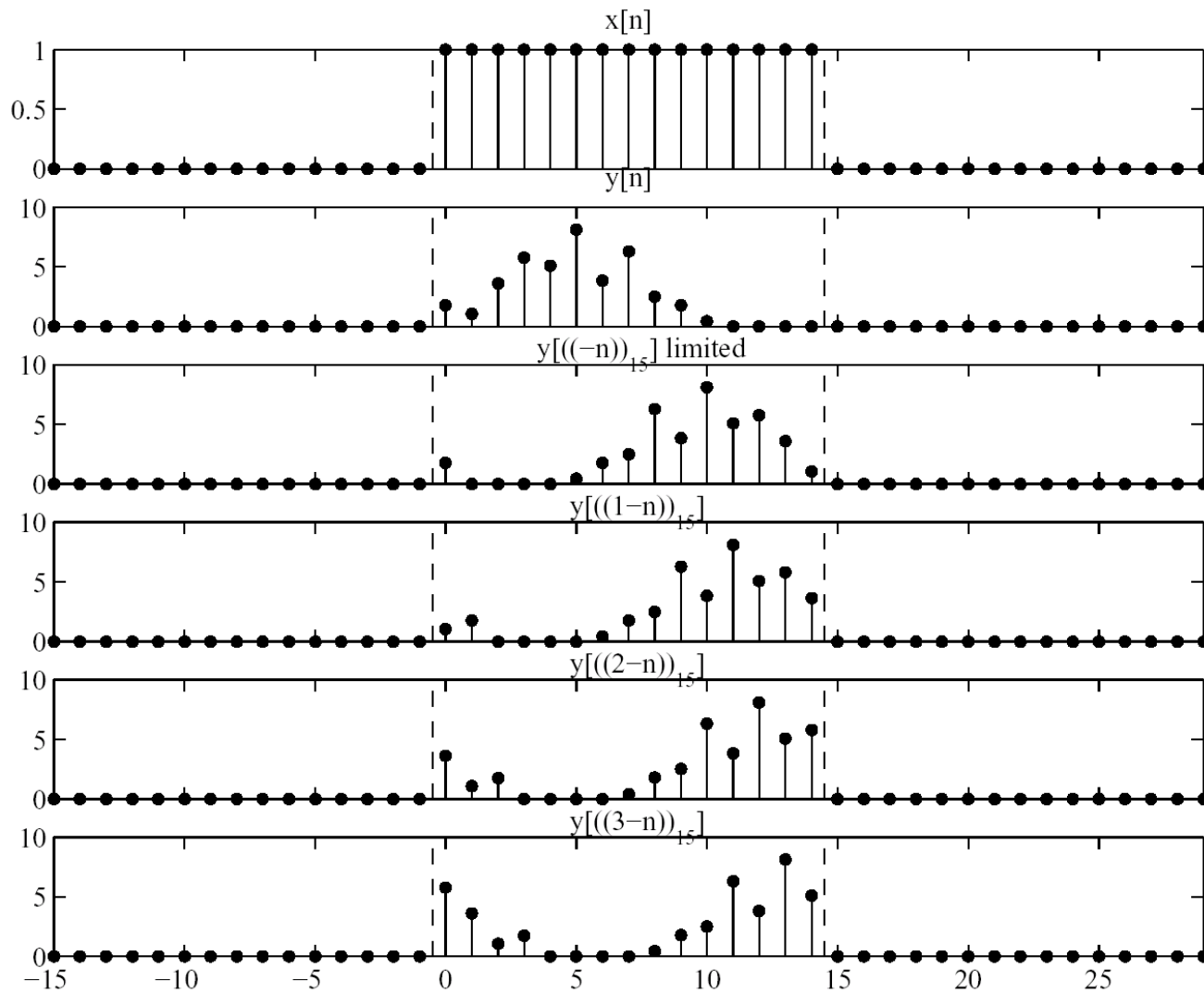
$$\sum_{m=0}^{N-1} x[m]y[((n-m))_N], \quad 0 \leq n \leq N-1 \xleftrightarrow{\mathcal{DFT}} X[k]Y[k]$$

i.e. *circular convolution*, denoted as $x[n] \circledast y[n]$

Example: Circular Convolution



Example: Circular Convolution



Example (continued)

illustration of the circular convolution process

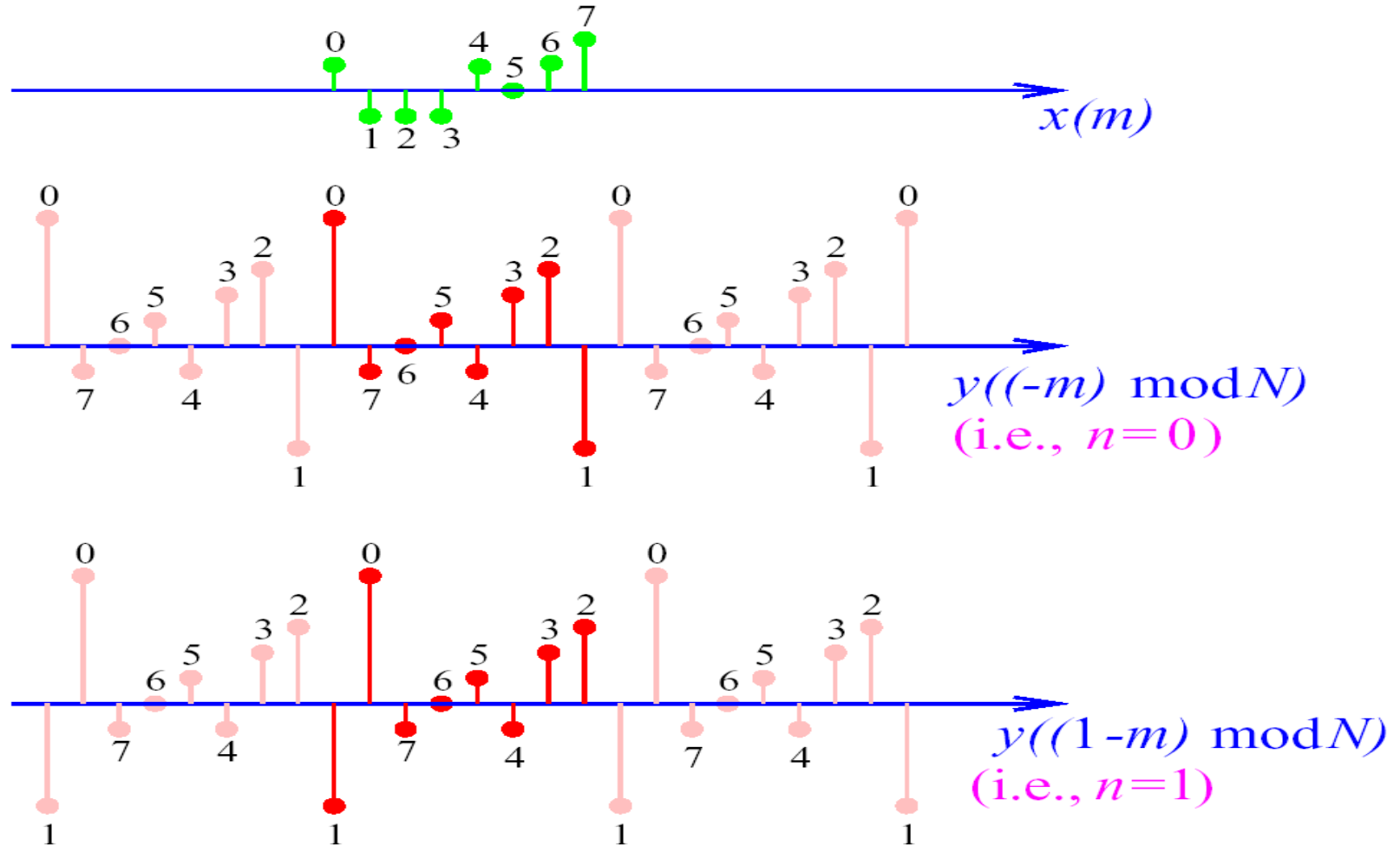
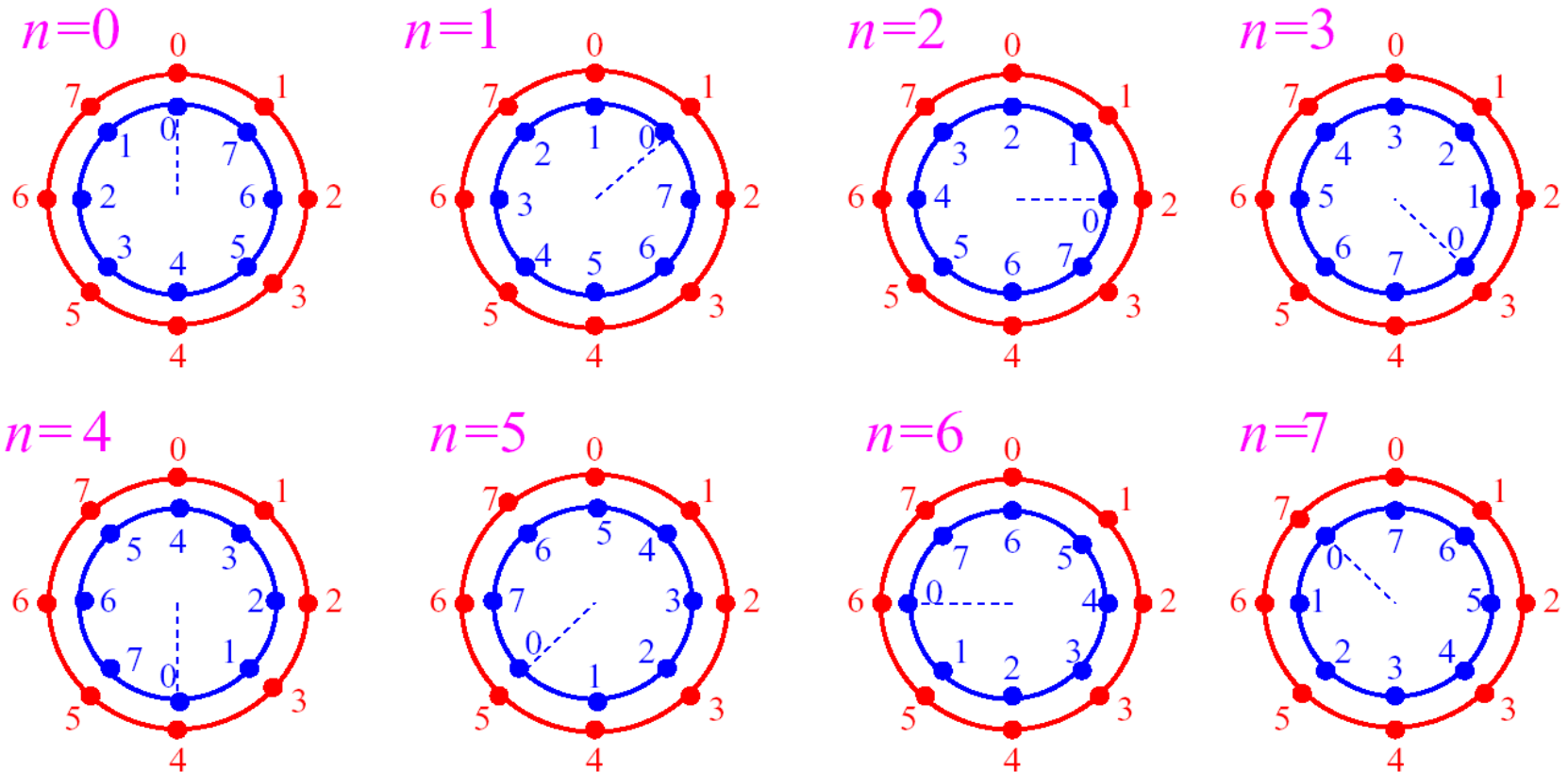


Illustration of circular convolution for $N = 8$:



- - $x(n)$ spread clockwise
- - $y(n)$ spread counterclockwise



$$\{x(n)\} = \{1, -1, -1, -1, 1, 0, 1, 2\}$$

$$\{y(n)\} = \{5, -4, 3, 2, -1, 1, 0, -1\}$$

and $\{z(n)\} = \{x(n)\} \circledast \{y(n)\}$. Then

$$\begin{aligned} z(0) &= x(0)y(0) + x(1)y(7) + x(2)y(6) + x(3)y(5) + x(4)y(4) \\ &\quad + x(5)y(3) + x(6)y(2) + x(7)y(1) = -1 \end{aligned}$$

$$\begin{aligned} z(1) &= x(0)y(1) + x(1)y(0) + x(2)y(7) + x(3)y(6) + x(4)y(5) \\ &\quad + x(5)y(4) + x(6)y(3) + x(7)y(2) = 1 \end{aligned}$$

$$\begin{aligned} z(2) &= x(0)y(2) + x(1)y(1) + x(2)y(0) + x(3)y(7) + x(4)y(6) \\ &\quad + x(5)y(5) + x(6)y(4) + x(7)y(3) = 6 \end{aligned}$$



$$\begin{aligned} z(3) &= x(0)y(3) + x(1)y(2) + x(2)y(1) + x(3)y(0) + x(4)y(7) \\ &\quad + x(5)y(6) + x(6)y(5) + x(7)y(4) = -4 \end{aligned}$$

$$\begin{aligned} z(4) &= x(0)y(4) + x(1)y(3) + x(2)y(2) + x(3)y(1) + x(4)y(0) \\ &\quad + x(5)y(7) + x(6)y(6) + x(7)y(5) = 5 \end{aligned}$$

$$\begin{aligned} z(5) &= x(0)y(5) + x(1)y(4) + x(2)y(3) + x(3)y(2) + x(4)y(1) \\ &\quad + x(5)y(0) + x(6)y(7) + x(7)y(6) = -8 \end{aligned}$$

$$\begin{aligned} z(6) &= x(0)y(6) + x(1)y(5) + x(2)y(4) + x(3)y(3) + x(4)y(2) \\ &\quad + x(5)y(1) + x(6)y(0) + x(7)y(7) = 4 \end{aligned}$$

$$\begin{aligned} z(7) &= x(0)y(7) + x(1)y(6) + x(2)y(5) + x(3)y(4) + x(4)y(3) \\ &\quad + x(5)y(2) + x(6)y(1) + x(7)y(0) = 7 \end{aligned}$$



$$\begin{aligned}\mathcal{DFT}\{\{x(n)\} \circledast \{y(n)\}\} &= \sum_{n=0}^{N-1} \underbrace{\left[\sum_{m=0}^{N-1} x(m) y((n-m) \bmod N) \right]}_{\{x(n)\} \circledast \{y(n)\}} W^{kn} \\ &= \sum_{m=0}^{N-1} \underbrace{\left[\sum_{n=0}^{N-1} y((n-m) \bmod N) W^{kn} \right]}_{Y(k) W^{km}} x(m) \\ &= Y(k) \underbrace{\sum_{m=0}^{N-1} x(m) W^{km}}_{X(k)} = X(k) Y(k)\end{aligned}$$



If $X(k) = \mathcal{DFT}\{x(n)\}$ and $Y(k) = \mathcal{DFT}\{y(n)\}$

then $\frac{1}{N}X(k) \otimes Y(k) = \mathcal{DFT}\{x(n)y(n)\}$

Also if $x(n) = \mathcal{DFT}^{-1}\{X(k)\}$ and $y(n) = \mathcal{DFT}^{-1}\{Y(k)\}$

then $x(n)y(n) = \mathcal{DFT}^{-1}\{\frac{1}{N}X(k) \otimes Y(k)\}$

Proof: similar to that for the circular convolution property